

Chapter C6: Making useful chemicals

Overview

This chapter considers the laboratory and large-scale production of useful chemicals.

Topic C6.1 begins with the laboratory synthesis of salts from acid reactions, and also looks at the characteristics of both acids and bases.

In Topic C6.2, the story moves on to study how chemists manage the rate of reaction when these reactions take place, in the context of managing conditions both in the laboratory and in industry. This topic gives the opportunity for a wide range of practical investigation and mathematical analysis of rates.

Topic C6.3 looks at reversible reactions, with particular emphasis on the large-scale production of ammonia.

Topic C6.4 develops all the ideas in the chapter together to look at both how the conditions and 'routes' for a chemical process are chosen by thinking about the economic and environmental issues surrounding the production of chemicals on a large scale. The manufacture of fertilisers is used as a context for considering how chemists reach decisions about the optimum processes for large-scale production of bulk chemicals.

Learning about Making useful Chemicals before GCSE (9–1)

From study at Key Stages 1 to 3 learners should:

- explain that some changes result in the formation of new materials, and that this kind of change is not usually reversible
- represent chemical reactions using formulae and using equations
- define acids and alkalis in terms of neutralisation reactions
- describe the pH scale for measuring acidity/alkalinity; and indicators
- recall reactions of acids with metals to produce a salt plus hydrogen and reactions of acids with alkalis to produce a salt plus water
- know what catalysts do.
- know about energy changes on changes of state (qualitative)
- know about exothermic and endothermic chemical reactions (qualitative).

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.

All other statements will be assessed in both Foundation and Higher Tier papers.

Learning about Making useful Chemicals at GCSE (9–1)

C6.1 What useful products can be made from acids?

Teaching and learning narrative	Assessable learning outcomes <i>Learners will be required to:</i>
Many products that we use every day are based on the chemistry of acid reactions. Products made using acids include cleaning products, pharmaceutical products and food additives. In addition, acids are made on an industrial scale to be used to make bulk chemicals such as fertilisers. Acids react in neutralisation reactions with metals, hydroxides and carbonates. All neutralisation reactions produce salts, which have a wide range of uses and can be made on an industrial scale. The strength of an acid depends on the degree of ionisation and hence the concentration of H^+ ions, which determines the reactivity of the acid. The pH of a solution is a measure of the concentration of H^+ ions in the solution. Strong acids ionise completely in solution, weak acids do not. Both strong and weak acids can be prepared at a range of different concentrations (i.e. different amounts of substance per unit volume). Weak acids and strong acids of the same concentration have different pH values. Weak acids are less reactive than strong acids of the same concentration (for example they react more slowly with metals and carbonates).	- recall that acids react with some metals and with carbonates and write equations predicting products from given reactants - describe practical procedures to make salts to include appropriate use of filtration, evaporation, crystallisation and drying <i>PAG7</i> - use the formulae of common ions to deduce the formula of a compound - recall that relative acidity and alkalinity are measured by pH including the use of universal indicator and pH meters use and explain the terms dilute and concentrated (amount of substance) and weak and strong (degree of ionisation) in relation to acids including differences in reactivity with metals and carbonates use the idea that as hydrogen ion concentration increases by a factor of ten the pH value of a solution decreases by one describe neutrality and relative acidity and alkalinity in terms of the effect of the concentration of hydrogen ions on the numerical value of pH (whole numbers only)

Linked learning opportunities

Specification links

- Writing formulae, balanced symbol and ionic equations.(C3.2)
- Concentration of solutions (C5.4)

Practical Work:

- Reactions of acids and preparation of salts.
- pH testing
- Investigating strong and weak acid reactivity.
- use of indicators to test strong and weak acids, making standard solutions using volumetric flasks.